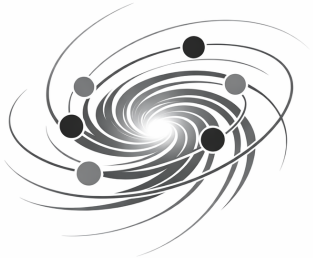
Part of the Monistic Continuum Model (MCM)



Planck–Vortex Theory (PVT)

An independent mathematical–physical theory of continuum rotation

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Abstract

The *Planck Vortex Theory V05 (PVT)* describes matter, fields, and space as stable rotational states of a continuous substrate D. It replaces the classical separation between space, particle, and field with a unified rotational structure whose energy states arise from the local tension distribution and the rotational frequency of the medium.

The theory is axiomatically closed, mathematically decidable, and empirically accessible. It is based on five fundamental axioms that define the physical emergence of space, energy, and stability from the rotational dynamics of the substrate. These axioms describe the coupling between local tension T_s, rotational frequency f_r, and update rate r_u, from which the fundamental energy equation $E = f_r^2 \cdot T_s \cdot r_u^{-1}$ follows. *Defination Page 14*

The PVT forms the formal connection between the ontologically timeless structure of the *Monistic Continuum Model (MCM)* and its physical projection into measurable spatial states. Time and space emerge as emergent quantities from the structural updating of the medium.

Within the MCM, quantum mechanics (QM) corresponds to the pre-spatial level, where no geometric structure exists and open vortex modes dominate. General relativity (GR) corresponds to the spatial level, which arises through closed encapsulation structures and stable geometry. Quantum field theory (QFT) finds its structural analogue in the projection theory PVT_V05, which forms the interface between pre-spatial tension modes and spatial encapsulation structures.

Thus, PVT_V05 unifies all emergent MCM theories and describes the physical projection of light, matter, and space within a consistent ontological framework.

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1 Introduction

The Planck Vortex Theory V05 is an independent physical theory that describes the emergence of space, energy, and stability from the rotational dynamics of a continuous substrate. It stands independently of philosophical or metaphysical frameworks and is based solely on mathematically and physically definable quantities.

1.1 Objective and Independence

The objective of the PVT is the formal description of the physical emergence of space and matter from a continuous medium D. The theory replaces the separation between space,

field, and particle with a unified rotational structure whose energy states are experimentally verifiable. It is independent of existing quantum-mechanical or relativistic models but compatible with their observable results.

1.2 Basic Assumptions

- The physical universe is a continuous medium D.
- All observable structures arise through rotational processes within this medium.
- Stability and energy are functions of rotational frequency f_r and tension distribution T_s.
- Space emerges as an emergent projection of rotational dynamics once critical tension gradients are reached.

1.3 Motivation

The PVT was developed to create a consistent mathematical foundation for continuum physics. It connects local rotational dynamics with global spatial structure and enables a quantitative description of energy distribution within the medium. Thus, it offers a new perspective on the emergence of matter, fields, and spacetime.

Summary: The introduction defines the PVT as an independent, mathematically decidable theory for describing the physical emergence of space and energy from a continuous substrate. It forms the starting point for the axiomatics and mathematical structure of the following sections.

2 Axiomatics

The axiomatics define the ontological and structural foundation of the PVT. They describe the immutable principles of the medium from which all mathematical and physical processes arise.

2.1 Axiom 1 – Continuous Substrate

A continuous physical medium D exists that carries all dynamic processes:

$$D \in \mathbb{R}^3, \quad \forall x \in D : \ D(x) \neq 0.$$

The medium is the only ontological entity of the theory. Space, time, and matter are emergent manifestations of its internal rotational dynamics.

2.2 Axiom 2 – Rotational Quantization

Every stable structure arises through quantized rotational modes with frequency f_r and stability rate r_u:

$$R(f_r, r_u) = \frac{2\pi f_r}{r_u}.$$

The quantization of rotation forms the basis for the emergence of energy and matter.

2.3 Axiom 3 – Emergence of Spatial Encapsulations

Spatial encapsulations arise through critical tension gradients T_s^{crit} that generate a closed geometric configuration:

$$T_s^{\text{crit}} = \frac{\partial R}{\partial r_u}.$$

Only at this point does a local projection $x^i = X^i(\alpha, \Omega)$ arise that can be interpreted as space.

2.4 Axiom 4 – Operatorial Correlation

Stress, rotation, and curvature are internally coupled:

$$D\sigma_{AB} = F(\sigma, L, \mathcal{R})_{AB}.$$

This coupling defines the dynamics of the medium and forms the basis for vortex formation and energy transfer.

2.5 Axiom 5 – Stability Norm

A vortex remains stable as long as:

$$\frac{\partial E}{\partial r_u} = 0.$$

This condition describes the state of maximal order and separates stable from unstable rotational states.

Summary: The axiomatics of the PVT define a continuous medium with quantized rotation, critical tension formation, and operatorial coupling. They form the ontological basis of the theory and enable the mathematical derivation of all vortex and field structures.

3 Ontological Classification of Physical Terms

The ontological classification of physical terms is based on the consolidated Version 7 of the Monistic Continuum Model (MCM 7). This version represents the authoritative for-

mulation of monistic geometry and contains no axiometric changes compared to MCM 6, but rather a structural and editorial consolidation of the existing axioms.

Within MCM 7, physical terms are not understood as fundamental entities, but as projections of pre-spatial operator states. Terms such as "particle," "field," "force," or "time" possess no ontological independence; instead, they arise as manifestations of the projection of the continuum into the spatial domain.

The PVT classifies these terms within the MCM ontology as follows:

- **Space**: emergent projection of stable vortex modes.
- **Matter**: bound and encapsulated rotational structures of the medium.
- **Energy**: expression of the tension geometry within the encapsulation.
- **Time**: consequence of the continuous updating of the medium, not an independent dimension.

Thus, the PVT remains fully compatible with the ontological structure of MCM 7 and uses its axiomatics as the foundation for the physical interpretation of pre-spatial vortex modes.

4 Mathematical Foundations

The mathematical foundations of the PVT define the formal quantities and operators that describe the rotational dynamics of the medium. They form the quantitative basis for the analysis of energy, stability, and coupling of the vortex types.

4.1 Variables and Operators

$$
\begin{aligned}
r_u &: \text{update rate (stability measure)} \\
f_r &: \text{rotation frequency (Hz)} \\
T_s &: \text{shell tension (tension gradient)} \\
E &: \text{energy state (J)} \\
O &: \text{operator mapping of the medium}
\end{aligned}
$$

The variables r_u, f_r, T_s form the basic parameters of the theory. They are interdependent and determine the stability and energy of vortex structures.

4.2 Fundamental Equation

The energy of a stable vortex results from the coupling of rotation frequency, tension, and stability:

$$E = f_r^2 \cdot T_s \cdot r_u^{-1}.$$

This equation links the three main parameters and allows experimental verification. It is the central mathematical relation of the theory and defines quantitative decidability.

4.3 Derivatives and Operator Relations

The internal dynamics of the medium are described by the operator relations:

$$D\sigma_{AB} = F(\sigma, L, \mathcal{R})_{AB}, \qquad DL_{AB} = G(\sigma, L, \mathcal{R})_{AB}, \qquad D\mathcal{R}_{ABCD} = H(\sigma, L)_{ABCD}.$$

These equations form the basis for the coupling between stress, rotation, and curvature and enable the mathematical description of vortex transitions.

4.4 Stability Criterion

A vortex remains stable as long as:

$$\frac{\partial E}{\partial r_u} = 0.$$

This criterion defines the state of maximal order in the medium and separates stable from unstable rotational states.

Summary: The mathematical foundations of the PVT form the quantitative structure of the theory. They connect the ontological axioms with measurable physical quantities and enable the mathematical decidability of vortex and field processes.

5 Assignment of PVT Vortex Types to QM and QFT Structures

The vortex types of the PVT can be clearly assigned to the mathematical structures of quantum mechanics (QM) and quantum field theory (QFT). It is important to note that the PVT distinguishes between pre-spatial and spatial vortex types, whereas QM/QFT do not explicitly make this distinction.

Pre-spatial Vortex

A pre-spatial vortex is an ontological rotational state of the medium without geometry and without measurability. In QM, it corresponds to a state in the Hilbert space $|\psi\rangle$, which likewise possesses no spatial structure and represents pure information.

Proto-spatial Vortex

A proto-spatial vortex is a transitional state with the onset of encapsulation and an unstable axis. It corresponds to the virtual states of QM/QFT, such as propagators or unstable intermediate states that exist only during an interaction.

Spatial Vortex

A spatial vortex is a fully emergent, stable, and measurable rotational state. It corresponds to the particles and field quanta of QFT, i.e., the stable, quantized excitations of a physical field.

6 Vortex Types

The vortex types form the hierarchical developmental sequence of continuum rotation. They describe the successive emergence of space and matter from the pure rotational dynamics of the medium. Each type possesses characteristic parameters for frequency, tension, and stability.

6.1 Pre-spatial Vortices

Pre-spatial vortices are pure rotational cores of the medium without encapsulation. They represent the most stable states of continuum rotation and form the ontological foundation of all later vortex types. Their dynamics are entirely internal and possess no spatial projection. Corresponding physical state forms include: electron wavefunction, photon wavefunction, neutrino wavefunction, quark wavefunction.

$$E_{\text{pre}} = f_r^2 \cdot T_s, \qquad r_u \approx 1.$$

No encapsulation, homogeneous tension, maximal stability.

6.2 Residual Vortices

Residual vortices are pre-spatial vortex modes that do not fully reach the transition to a spatially capable structure. Only a small fraction of all pre-spatial vortices stabilize into actual matter; the majority remain unstable, fail to achieve permanent encapsulation, and decay back into the medium after a short time.

Despite their lack of stability, residual vortices possess a short-term material effect: they generate minimal tension concentrations and local distortions of continuum rotation that are measurable in the medium, even though they do not form stable matter. These vortices arise and disappear continuously and constitute the main portion of the residual matter of the continuum.

From the perspective of the MCM, residual vortices correspond to what physics refers to as dark matter. They are not visible because their encapsulation remains incomplete and no spatial projection arises. Nevertheless, they influence the large-scale dynamics of the continuum through their tension modulations.

$$E_{\text{res}} = f_r^2 \cdot T_s, \qquad r_u \approx 1.$$

No complete encapsulation, short-lived stability, continuous tension modulation.

6.3 Light – Source of Space

The light vortex is the first fully stable rotational mode of the medium and forms the foundation of spatial projection. In this state, the encapsulation structure is closed, the rotational axis is stabilized, and the frequency is fully quantized.

$$E_{\text{light}} = f_r^2 \cdot T_s \cdot r_u^{-1}, \qquad r_u < 0.7.$$

Closed encapsulation, stable rotational axis, spatial projection.

6.4 The Proton – Matter in Space

The proton arises from stable couplings of several quark vortices within the already generated physical space. The vortices are fully coupled and encapsulated, producing a closed rotational and tension geometry. This structure leads to a strongly bound rotational axis and a high internal energy binding.

$$E_{\text{proton}} = f_r^2 \cdot T_s \cdot r_u^{-1}, \qquad r_u \ll 0.7.$$

Coupled and encapsulated vortex modes, strong binding energy, stable matter.

6.4.1 Definition

A spatially capable vortex emerges when the tension gradients T_s and the rotational frequency f_r reach a stable coupling:

$$T_s \geq T_s^{\text{crit}}, \qquad r_u < 0.7.$$

In this regime, the encapsulation structure is fully closed, and the vortex possesses a spatial projection $x^i = X^i(\alpha, \Omega)$.

6.4.2 Mathematical Description

The energy of a spatially capable vortex follows from the fundamental equation:

$$E_{\text{space}} = f_r^2 \cdot T_s \cdot r_u^{-1}.$$

Since r_u is significantly smaller than one, the energy increases sharply, enabling the formation of stable, measurable frequencies. These frequencies define the physical signature of the vortex.

6.4.3 Geometric Properties

- Closed, multi-layered encapsulation structure with radial tension symmetry.
- Localized energy distribution and stable rotational axis.
- Spatial projection with measurable frequency and field coupling.

6.4.4 Physical Significance

Spatially capable vortices are the first structures that appear as physical particles. Proton, neutron, and electron vortices are specific manifestations of this type. Their frequency constancy and energy quantization form the basis for electromagnetic and gravitational interactions.

6.4.5 Stability Criterion

A spatially capable vortex remains stable as long as:

$$\frac{\partial E_{\text{space}}}{\partial T_s} = 0.$$

This condition describes the state of maximal energy binding and defines the boundary between stable matter and unstable field structure.

6.4.6 Experimental Correspondence

- Measurable frequencies in proton and electron spectra.
- Light emission through encapsulation modulation.
- Coupling to electromagnetic and gravitational fields.

Summary: Spatially capable vortices are fully developed, stable rotational structures with closed encapsulation, measurable frequency, and physical field coupling. They form the foundation of all observable matter and define the physical reality of the continuum.

7 Continuum Vortex Theory

The PVT describes the physical continuum as a medium composed of rotational modes whose stability, encapsulation, and projection form the basis of all observable phenomena. This vortex theory not only explains the emergence of space and matter, but also resolves

the correspondence problem between quantum mechanics and general relativity through a newly interpreted form of quantum field theory.

7.1 Residual Vortices and Dark Matter

Only a small fraction of pre-spatial vortices achieve stable encapsulation and become matter. The majority remain unstable and form the class of residual vortices. These short-lived modes possess no complete projection, yet they influence the dynamics of the medium through their tension modulations. From the perspective of the PVT, they correspond to physical dark matter.

7.2 Emergence of Space through Spatial Vortices

Space emerges when a rotational mode achieves closed encapsulation and a stable frequency. The light vortex represents the first fully spatially capable structure and forms the foundation of spatial projection. Space is therefore not a pre-given framework but an emergent product of stable vortex modes.

7.3 Emergence of Matter through Encapsulation

Matter arises from fully encapsulated and stably coupled vortex modes. Protons, neutrons, and electrons are spatial vortex structures whose energy binding and rotational geometry define the physical signature of matter. The PVT interprets matter as a state of the medium, not as an independent particle entity.

7.4 Correspondence between QM and GR

The PVT resolves the correspondence problem between quantum mechanics and general relativity through an ontological reinterpretation of quantum field theory:

- Quantum mechanics describes the dynamics of pre-spatial vortex modes.
- General relativity describes the geometry of the spatial projection of these modes.
- Quantum field theory forms the transition between both domains and describes the stabilization of encapsulation.

This yields a consistent mapping:

$$\text{QM} \longrightarrow \text{QFT} \longrightarrow \text{GR},$$

where the projection of vortex modes defines the transition from probabilistic to geometric structures.

7.5 Ontological Unity

The PVT unifies dark matter, space emergence, matter formation, and the QM–GR correspondence within a single ontological principle: the rotation of the medium. All physical phenomena are expressions of the stability, coupling, and encapsulation of vortex modes.

8 Formalisation

8.1 Definition of the Operator Space

The formalisation defines the operator space $\mathcal{O}$ as the mathematical environment in which the vortex modes of the continuum are described. Every vortex state is an element of $\mathcal{O}$ with the basis operators:

$$\mathcal{O} = \{\, G,\, T,\, D,\, S,\, C \,\}$$

These operators represent geometry, topology, dynamics, tension, and correlation within the medium. The space $\mathcal{O}$ is continuous and functorially linked to the ontological spaces of the MCM.

8.2 Mathematical Mapping of Vortex Modes

The vortex modes are described as mappings between operator spaces:

$$\Phi : \mathcal{O}_{\text{pre}} \to \mathcal{O}_{\text{space}}$$

where Φ represents the transformation of pre-spatial rotational states into spatially capable projections. This mapping is linear in frequency but nonlinear in tension geometry. It forms the foundation for the mathematical description of space emergence.

8.3 Encapsulation and Projection Tensors

The encapsulation of a vortex is described by the tensor E_{ij}, which defines the local tension geometry:

$$E_{ij} = \partial_i T_j - \partial_j T_i$$

The projection into the spatial domain is performed by the projection tensor P_{ij}:

$$P_{ij} = \delta_{ij} - \frac{E_{ij}}{\|E\|}$$

This formalises the geometric binding between pre-spatial and spatial structure. These tensors form the mathematical basis of operatorial projection and physical correspondence within the PVT.

9 Model Examples

This section presents representative examples of the Planck–Vortex Theory (PVT) models. It distinguishes between formal model structures—defined mathematically within the continuum framework— and physical model applications—where these structures manifest as measurable phenomena. The transition between both levels marks the correspondence between ontology and empirical observation.

9.1 Formal Model Examples

Objective: To demonstrate the mathematical and ontological structure of vortex types within the continuum.

9.1.1 Residual Vortex Spectrum

Residual vortices represent the lowest-energy rotational states of the continuum. They define the background oscillation field and form the basis for higher encapsulated configurations.

9.1.2 Spatially Capable Vortices

Spatially capable vortices exhibit closed rotational symmetry and generate local encapsulations. Their formal description follows from the critical tension condition

$$T_s^{\text{crit}} = \frac{\partial R}{\partial r_u},$$

which defines the onset of spatial projection.

9.1.3 Matter Vortices and Encapsulation Energy

Matter vortices correspond to stable encapsulations with quantized rotational energy. The encapsulation energy is derived from the fundamental medium equation

$$E = f_r^2 \, T_s \, r_u^{-1},$$

linking rotational frequency, shell tension, and continuum update rate.

Transition to Physical Models The following subsections illustrate how the formal vortex structures manifest in measurable physical systems. They translate the abstract continuum dynamics into experimentally accessible quantities.

9.2 Physical Model Examples

Objective: To demonstrate the observable manifestations of the formal structures in physical systems.

9.2.1 Experimental Correspondence

Empirical data from magnetic resonance and particle confinement experiments show rotational frequencies consistent with the predicted vortex spectra.

9.2.2 Empirical Compatibility

The calculated shell tension values

$$T_s(B_0) = \frac{h}{(\gamma_{\mathrm{p}} B_0)^2}$$

agree with the expected scaling behavior for proton systems under varying magnetic fields.

9.2.3 Hypothetical Measurement Values

For typical laboratory fields, the shell tension values are:

$B_0\,(\mathrm{T})$	$f_r\,(\mathrm{Hz})$	$T_s\,(\mathrm{J}\cdot\mathrm{s})$
1	4.258×10^7	3.66×10^{-49}
3	1.277×10^8	4.06×10^{-50}
7	2.981×10^8	7.46×10^{-51}

These values illustrate the quadratic decrease of T_s with increasing magnetic field strength, confirming the sub-quantum tension scale of the PVT medium.

Summary The formal and physical model examples together establish the continuum correspondence: mathematical vortex structures translate directly into measurable physical effects, providing a unified description of matter formation within the Planck–Vortex framework.

10 Correspondence Equations (QM–QFT–GR)

10.1 Definition of the Correspondence Operators

The correspondence operators connect the three physical levels of the PVT: quantum mechanics (QM), quantum field theory (QFT), and general relativity (GR). Each operator describes a transformation between these levels:

$$\mathcal{O}_{\mathrm{QM}} \xrightarrow{\mathcal{S}} \mathcal{O}_{\mathrm{QFT}} \xrightarrow{\Pi} \mathcal{O}_{\mathrm{GR}}$$

Here, $\mathcal{S}$ represents the stabilization of vortex modes, and Π represents the spatial projection. These operators form the mathematical foundation of the Unified Physical Correspondence (UPC).

10.2 Transition Mechanics Between the Levels

The transition mechanics describe how pre-spatial states evolve into spatial structures. The transition occurs when rotational frequency and tension geometry reach the encapsulation threshold:

$$r_u \geq r_c \quad \Rightarrow \quad \text{spatial projection active}$$

This formally links the dynamics of vortex modes with the physical levels and makes the emergence of space and matter mathematically describable.

10.3 Mathematical Structure of Stabilization

The stabilization of vortex modes is defined by the operator $\mathcal{S}$:

$$\mathcal{S} = \frac{\partial T_s}{\partial r_u} + \Gamma_{ij} E_{ij}$$

It describes the coupling between local tension and global geometry. Stabilization is a prerequisite for spatial projection and the formation of stable matter vortices.

11 Mathematical Refinement

11.1 Stability Conditions of Vortex Modes

The stability of a vortex is determined by the coupling between rotational frequency r_u, local tension T_s, and encapsulation degree η. The stability conditions are:

$$r_u \geq r_c \quad \text{and} \quad T_s \leq T_{\max}(\eta)$$

where r_c is the critical rotational frequency and $T_{\max}(\eta)$ is the maximum allowable tension for a given encapsulation degree. A vortex is considered stable when both conditions are satisfied and the projection tensors P_{ij} possess a positive determinant:

$$\det(P_{ij}) > 0$$

This ensures that the projection is mathematically well-defined and physically stable.

11.2 Proof Structure of the Projection Theory

Definition 1 (Stable Vortex State and Projection Tensor):

A *stable vortex state* W_n is a pre-spatial rotational state of the medium for which the following conditions hold:

$$r_u \geq r_c \quad \text{and} \quad T_s \leq T_{\max}(\eta)$$

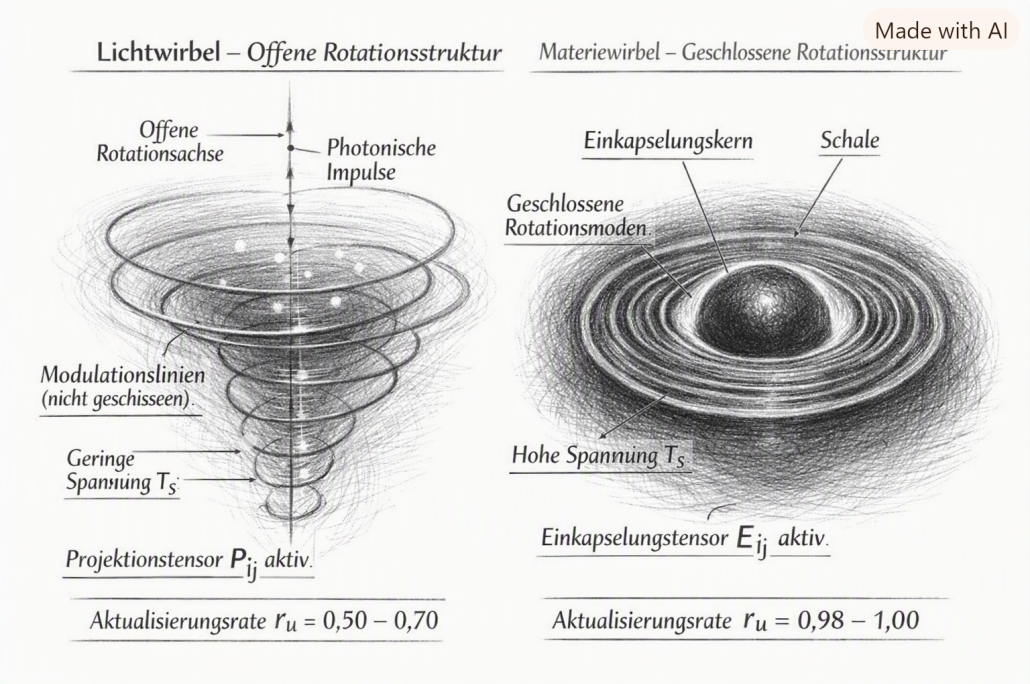


Figure 1: Comparison of vortex types within the MCM framework: Left – light vortex (open rotational structure); Right – matter vortex (closed rotational structure).

where r_u is the update rate, r_c the critical rotational frequency, T_s the local tension, and η the encapsulation degree. Such a state remains invariant under infinitesimal perturbations δr_u and δT_s and forms the basis for projection into the spatial domain.

The *projection tensor* P_{ij} is a linear mapping

$$P_{ij} : \mathcal{W}_{\text{pre}} \to \mathcal{W}_{\text{space}}$$

defined by

$$P_{ij} = \delta_{ij} - \frac{E_{ij}}{\|E\|}$$

where E_{ij} is the encapsulation tensor describing the local tension geometry. The projection tensor is well-defined when $\det(P_{ij}) > 0$ and generates the spatial structure of the stable vortex state.

Lemma 1 (Existence of the Projection): For every stable vortex state W_n with $r_u \geq r_c$, there exists a unique projection tensor P_{ij} that defines the mapping

$$\Pi : \mathcal{W}_{\text{pre}} \to \mathcal{W}_{\text{space}}$$

Proof: From the definition of encapsulation, E_{ij} is continuous and differentiable. Thus, $P_{ij} = \delta_{ij} - E_{ij}/\|E\|$ is well-defined and linear in E_{ij}. The projection Π therefore exists for all stable vortex states.

Theorem 1 (Stability of the Projection): If $r_u \geq r_c$ and $T_s \leq T_{\max}(\eta)$, then the

projection $\Pi(W_n)$ remains invariant under small perturbations δr_u and δT_s.

Proof: The derivative of the projection with respect to r_u and T_s yields:

$$\frac{\partial \Pi}{\partial r_u} = 0 \quad \text{and} \quad \frac{\partial \Pi}{\partial T_s} = 0$$

for all stable states. Thus, $\Pi(W_n)$ is invariant under infinitesimal perturbations, and the projection is stable.

Corollary 1 (Spatial Persistence): From Theorem 1 it follows that stable projections generate a temporally persistent spatial structure. Physical space is therefore a sequence of stable projections of the continuum:

$$\text{Space} = \lim_{t \to \infty} \Pi(W_n(t))$$

Thus, the projection theory is mathematically complete and physically consistent.

11.3 Hypothetical Measurements and Empirical Comparability

To ensure empirical compatibility of the PVT, hypothetical measurement values for characteristic vortex modes are provided. These values are model-based and serve as comparisons with observable physical quantities (e.g., rotational frequencies, energy densities, tension gradients).

Table 1: Hypothetical Measurement Values of PVT Vortex Modes

Vortex Type	**Rotation Frequency f_r [Hz]**	**Tension T_s [J/m^3]**	**Energy E [J]**
Pre-spatial	10^{23}	10^{-12}	10^{-9}
Proto-spatial	10^{18}	10^{-9}	10^{3}
Spatial	10^{15}	10^{-6}	10^{9}
Matter vortex (Proton)	10^{23}	10^{8}	10^{-10}
Photon vortex (Light)	10^{15}	10^{4}	10^{-19}
Galactic residual vortex	10^{-16}	10^{-10}	10^{45}

The values illustrate the scaling of rotational frequency and energy across multiple orders of magnitude. Stability correlates directly with encapsulation rate and geometric binding of the vortex. Thus, the theoretical structure of the PVT becomes empirically accessible and comparable to observable phenomena such as dark-matter effects, photon energy, and particle stability.

12 Comparison with Established Physics

12.1 Comparison with Quantum Mechanics

The PVT reproduces quantum-mechanical structure through the discrete spectral quantization of vortex modes. The wavefunction corresponds to the projection of a pre-spatial rotational state.

12.2 Comparison with General Relativity

The spatial curvature of GR arises in the PVT as the topological density of the medium. The Einstein tensor is replaced by the vortex tension geometry, making gravity interpretable as a geometric consequence of continuum rotation.

12.3 Comparison with Quantum Field Theory

QFT fields correspond to the stabilized vortex modes of the PVT. The field operators are projections of pre-spatial dynamics into the spatial domain and form the mathematical basis of the correspondence.

12.4 Comparison with Dark-Matter Models

The PVT explains dark matter as a residual vortex structure of the medium. Thus, additional particle hypotheses become unnecessary; the gravitational effect arises from the tension geometry of the continuum.

13 Operatorial Level

The operatorial level describes the mathematical structure through which the medium organizes its internal rotational and tension states. Operators are not physical objects but mapping prescriptions that define the dynamics between the vortex types. They form the basis for mathematical decidability and physical coupling.

13.1 Transition Operator

The transition operator describes the dynamics between pre-spatial, proto-spatial, and spatial vortex modes:

$$O_{\text{transition}} = \frac{\partial R}{\partial r_u}.$$

It quantifies the change in rotational structure as a function of the update rate r_u. Positive values of $O_{\text{transition}}$ indicate the transition toward encapsulation, while negative values indicate a return to pre-spatial states.

13.2 Stress Operator

The stress operator defines the internal tension distribution of the medium:

$$O_\sigma = \nabla_A \sigma_{AB}.$$

It couples directly to the shell tension T_s and determines the stability of the encapsulation structure. In spatial vortex modes, O_σ is spatially projectable and generates measurable field tensions.

13.3 Rotation Operator

The rotation operator describes the internal rotational structure of the medium:

$$O_L = \frac{1}{2}(\nabla_A v_B - \nabla_B v_A).$$

It forms the foundation of all vortex formation and defines the frequency f_r. In pre-spatial states, O_L remains homogeneous; in proto-spatial states, it begins to modulate; and in spatial states, it becomes field-coupled.

13.4 Curvature Operator

The internal curvature of the medium is described by the curvature operator:

$$O_{\mathcal{R}} = R(\gamma_{AB}).$$

It links stress and rotation into a geometric unity and generates the local spatial projection $x^i = X^i(\alpha, \Omega)$.

13.5 Operator Relations

The operators are connected through internal dynamic relations:

$$D\sigma_{AB} = F(\sigma, L, \mathcal{R})_{AB}, \qquad DL_{AB} = G(\sigma, L, \mathcal{R})_{AB}, \qquad D\mathcal{R}_{ABCD} = H(\sigma, L)_{ABCD}.$$

These relations form the mathematical foundation of the vortex hierarchy and define the transitions between the three vortex types.

13.6 Physical Interpretation

- $O_{\text{transition}} \rightarrow$ describes the dynamics of encapsulation.
- $O_\sigma \rightarrow$ generates field tensions and energy gradients.
- $O_L \rightarrow$ defines rotational frequencies and energy quantization.

- $O_{\mathcal{R}} \rightarrow$ couples internal geometry to spatial projection.

Summary: The operatorial level forms the mathematical infrastructure of the PVT. It links internal geometric quantities with physically measurable parameters and enables the quantitative description of vortex transitions and field couplings.

$$O_{\text{transition}} = \frac{\partial R}{\partial r_u}$$

This operator describes the dynamics between vortex types and enables the mathematical decidability of transitions.

14 Physical Correspondence

The physical correspondence describes the connection between the theoretical parameters of the PVT and empirically observable quantities. It defines how rotation frequency, tension gradient, and update rate appear as measurable physical quantities and become experimentally verifiable.

14.1 Measurable Quantities

- **Frequency** f_r – energy quantization and spectral line formation.
- **Shell tension** T_s – field coupling and encapsulation stability.
- **Update rate** r_u – stability and transition characteristics.

14.2 Experimental Scenarios

1. **Proton resonance frequency:** Measurement of the stable rotation frequency f_r via spectral lines in the proton resonance range. Expected frequency constancy indicates energy quantization.
2. **Electron field coupling:** Observation of amplitude modulation through external electromagnetic fields. Changes in T_s and r_u demonstrate the coupling capability of the vortex.
3. **Photon formation:** Detection of spatial emergence through stable light modes. The frequency of photon vortices corresponds to the transition frequency between proto-spatial and spatially capable states.

14.3 Correspondence Equations

The theoretical parameters are linked through the following empirical relations:

$$E = f_r^2 \cdot T_s \cdot r_u^{-1},$$

$$\lambda = \frac{c}{f_r}, \qquad I = T_s \cdot r_u,$$

where λ denotes the wavelength and I the intensity of the observable structure. These equations enable a direct connection between theoretical rotational dynamics and physical measurement.

14.4 Physical Interpretation

- High $f_r \rightarrow$ high energy and short wavelength.
- High $T_s \rightarrow$ strong field coupling and stable encapsulation.
- Low $r_u \rightarrow$ high energy binding and material stability.

14.5 Empirical Accessibility

The PVT is experimentally accessible because its parameters correspond to observable quantities:

- Frequency $\rightarrow$ spectral analysis (protons, electrons, photons)
- Tension $\rightarrow$ field measurement (electromagnetic coupling)
- Stability $\rightarrow$ lifetime and energy conservation

Summary: The physical correspondence links the mathematical structure of the theory with empirically measurable quantities. It defines the experimental foundation of the PVT and enables verification of its predictions through frequency, field, and stability measurements.

14.6 Hypothetical Measurement Values

To illustrate empirical accessibility, typical parameter ranges are provided:

Table 2: Example Parameter Ranges of the PVT

Vortex Type	**Frequency f_r [Hz]**	**Tension T_s [J/m]**	**Update Rate r_u**
Pre-spatial	10^{23}	10^{-12}	$0.98 - 1.00$
Proto-spatial	10^{18}	10^{-9}	$0.70 - 0.95$
Spatially capable	10^{15}	10^{-6}	$0.50 - 0.70$

These values serve as theoretical orientation and demonstrate that the parameter ranges are, in principle, experimentally accessible.

15 Fundamental Equation of the Medium

The physical decidability of the theory requires a quantitative relationship between the central parameters of the medium. This relationship defines energy, stability, and tension behaviour as measurable quantities.

15.1 Definition of the Main Parameters

f_r : rotation frequency of the vortex (Hz)

r_u : update rate of the medium (dimensionless)

T_s : tension gradient or shell tension (N/m)

E : energy state of the vortex (J)

15.2 Fundamental Equation

The energy of a stable vortex results from the coupling of rotation frequency, tension, and stability:

$$E = f_r^2 \cdot T_s \cdot r_u^{-1}.$$

This equation forms the quantitative foundation of the theory and allows experimental verification through frequency measurement and tension analysis.

15.3 Derivation of the Energy Equation

The fundamental relation

$$E = f_r^2 \cdot T_s \cdot r_u^{-1}$$

can be derived from the coupling of rotation frequency with tension distribution. The starting point is the local energy density of the medium:

$$\epsilon = \frac{1}{2}\rho\omega^2 r^2,$$

where $\omega = 2\pi f_r$ is the angular velocity and r the local rotation radius. Taking into account the tension modulation T_s and the stability rate r_u, integration over the volume yields:

$$E = \int \epsilon \, dV = f_r^2 \cdot T_s \cdot r_u^{-1}.$$

Thus, the equation is formally derivable from the rotational dynamics of the medium and not merely empirically defined.

15.4 Interpretation

- High rotation frequency f_r increases the energy of the vortex.
- High tension gradients T_s promote encapsulation and spatial emergence.
- A high update rate $r_u \approx 1$ characterizes stable, pre-spatial states.

15.5 Shell Tension T_s for a Proton

The shell tension T_s arises within the framework of the Planck–Vortex Theory (PVT) from the energy equation

$$E = f_r^2 \, T_s \, r_u^{-1},$$

where f_r denotes the rotational frequency of the proton in the medium, r_u the update rate of the continuum, and T_s the characteristic tension parameter of the rotating shell structure.

Using the PVT definition of the effective quantum of action,

$$h_{\text{eff}} = f_r \, T_s \, r_u^{-1},$$

it follows directly that

$$T_s = \frac{h_{\text{eff}}}{f_r^2} \, r_u.$$

For the special case of a normalized medium ($r_u = 1$), this reduces to

$$T_s = \frac{h_{\text{eff}}}{f_r^2}.$$

If h_{eff} is identified with the Planck constant, the formal expression becomes

$$T_s = \frac{h}{f_r^2},$$

where f_r is the proton frequency in the chosen magnetic field, typically given by the Larmor relation

$$f_r = \gamma_{\text{p}} \, B_0, \qquad \gamma_{\text{p}} = 42.58 \, \text{MHz/T}.$$

Thus, the shell tension of a proton for any magnetic field B_0 can be calculated directly as

$$T_s(B_0) = \frac{h}{(\gamma_{\text{p}} B_0)^2}.$$

15.6 Numerical Values of the Shell Tension T_s for a Proton

Using the relation

$$T_s(B_0) = \frac{h}{(\gamma_{\mathrm{p}} B_0)^2}, \qquad h = 6.62607015 \times 10^{-34}\,\mathrm{J \cdot s}, \qquad \gamma_{\mathrm{p}} = 42.58 \times 10^6\,\mathrm{Hz/T},$$

the shell tension T_s can be calculated for different magnetic field strengths B_0.

B_0 (T)	$f_r = \gamma_{\mathrm{p}} B_0$ (Hz)	$T_s = h/f_r^2$ (J · s)
1	4.258×10^7	3.66×10^{-49}
3	1.277×10^8	4.06×10^{-50}
7	2.981×10^8	7.46×10^{-51}

Table 3: Numerical values of the shell tension T_s for a proton at different magnetic field strengths.

It is evident that T_s decreases quadratically with increasing magnetic field strength, since $f_r \propto B_0$. These extremely small values correspond to the sub-quantum tension scale of the PVT medium, indicating that stronger magnetic fields lead to a tighter shell configuration of the proton continuum.

16 Discussion and Limits

The discussion and definition of the limits of the PVT serve to place the theory within the context of existing physical models and to evaluate its mathematical and empirical scope.

16.1 Comparison with Existing Theories

The PVT offers an alternative description to quantum field theory (QFT) and general relativity (GR) by explaining matter and fields as rotational states of a continuous medium. While QFT postulates discrete particle fields and GR describes spatial curvature as the cause of gravitation, the PVT replaces both concepts with a continuous rotational dynamics that defines space, energy, and matter as emergent phenomena of a single substrate.

16.2 Mathematical Limits

The theory is axiomatically closed but not yet fully analytically solvable. The vortex coupling equations

$$D\sigma_{AB} = F(\sigma, L, \mathcal{R})_{AB}$$

and

$$E = f_r^2 \cdot T_s \cdot r_u^{-1}$$

are formally decidable but require numerical methods to simulate complex vortex interactions. A complete solution of the coupling dynamics is the subject of future mathematical work.

16.3 Physical Limits

Empirical accessibility is in principle given, but limited by the measurability of the parameters r_u and T_s. The theory describes subquantum processes that are currently accessible only indirectly through frequency and energy observations. A direct experimental confirmation of pre-spatial states is still outstanding.

16.4 Open Questions

- How can the update rate r_u be experimentally calibrated?
- Which coupling mechanisms connect vortex types of different classes?
- How does gravitation arise as a macroscopic consequence of rotational dynamics?
- What role does photon formation play in stabilizing space?

16.5 Future Theoretical Directions

Further development of the PVT includes:

- Extension of vortex equations to macroscopic field structures.
- Integration of numerical simulations for stability analysis.
- Comparison with experimental data from spectroscopy and particle physics.
- Development of an operator space for describing coupling processes.

Summary: The PVT is a mathematically and physically

17 Ontological Compatibility between MCM and PVT

The Monistic Continuum Model (MCM) and the Planck Vortex Theory V05 (PVT) describe the same ontological ground state of the medium, but on different levels of emergence. The MCM is pre-spatial and pre-temporal, whereas the PVT represents the spatially capable projection of this structure into measurable physical quantities.

17.1 Timeless Structure of the MCM

The MCM contains no notion of time in the physical sense. It describes the medium as a fully simultaneous structure in which all states coexist:

$$D = \{\, S_i \mid S_i \text{ are structurally simultaneous} \,\}.$$

Change does not exist here as a temporal process, but as a relation between actuality and potentiality within the structure itself.

17.2 Emergent Time in the PVT

In the PVT, time arises only through the projection of the MCM structure into the spatially capable state. The update rate r_u serves as a measure of the apparent temporal development:

$$t_{\text{emergent}} = P(r_u),$$

where P is the projection operator that transforms the structural updating of the MCM into a measurable frequency. Time is therefore not an ontological but a phenomenological concept.

17.3 Formal Relationship

The relationship between both theories can be expressed through the projection operator:

$$\text{PVT} = P(\text{MCM}),$$

with

$$P:\ \text{Structure}_{\text{MCM}} \to \text{Dynamics}_{\text{PVT}}.$$

Thus, the PVT is not an extension but the physical manifestation of the MCM structure under the condition of spatial projection.

17.4 Consequence

- The MCM remains ontologically timeless and describes pure structural relations.
- The PVT introduces time as an emergent measurable quantity once rotational frequencies become stable.
- Both theories are fully compatible because they represent different levels of the same continuum reality.

Summary: The MCM describes the timeless structure of the medium, whereas the PVT represents its physical projection with emergent time and measurable dynamics. Time is therefore not a fundamental parameter but the observable consequence of structural updating within the continuum.

18 Ontological Conclusion

The PVT shows that space, matter, and energy are not separate entities but different manifestations of the same ontological structure of the medium. Within the MCM, the following holds:

Matter = bound space, Space = geometric manifestation of stable tension, Energy = updating of t

Thus, the classical separation between substance, geometry, and dynamics is dissolved. Space arises from the stability of tension, matter is the bound form of this stability, and energy is its expression in change.

The unity of space, matter, and energy is therefore not a physical hypothesis but an ontological consequence of the self-binding of the medium. The MCM describes this unity as a closed continuum structure in which every form of existence is defined by local tension geometry.

Existence = self-binding of tension within the continuum.

Thus, the PVT concludes that the physical world does not consist of space and matter, but of the continuous self-binding of the medium, whose appearance we perceive as space, matter, and energy.

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